04 General Principles and Processes of Isolation of Elements-Metallurgy

Introduction:

The compound of a metal found in nature is called a **mineral**.
The minerals from which metal can be economically and conveniently extracted are called **ores**.
An ore is usually contaminated with earthy or undesired materials known as **gangue**.

Note:

All minerals are not ores but all ores are minerals. Ores may be classified mainly into following two classes.



(a) Native ores :
Silver, gold, platinum etc, occur as native ores.

(b) Combined ores :
They contain the metal in combined form.
(i) Oxidised ores : Oxide ores , Carbonate ores, Sulphate ores, Phosphate ores, Silicate ores.
(ii) Sulphurised ores : These ores consist of sulphides of metals like iron, lead, zinc, mercury etc.
(iii) Halide ores : These ores consist of halides of metals

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List of Ores and Their Names:

Sulphide Ore:

PbS	→	Galena
HgS	→	Cinnabar
ZnS	→	Zinc blende
Cu_2S	→	Copper glance
$CuFeS_2$	→	Copper Pyrites (Chalcopyrite)
FeS_2	→	Iron pyrites or Fool's gold
Ag_2S	→	Silver glance or Argentite
$Cu_2S.Ag_2S$	→	Copper silver glance
$Ag_2S.Sb_2S_3$	→	Pyrargyrite or Ruby Silver

Halide Ore:

NaCl	→	Rock Salt
KCl	→	Sylvine
CaF_2	→	Fluorspar
Na_3AlF_6	→	Cryolite
AgCl	→	Horn Silver
$KCl.MgCl_2.6H_2O$	→	Carnalite

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Oxide Ore:

Fe_2O_3	→	Haematite
Fe_3O_4	→	Magnetite
$Al_2O_x(OH)_{3-2x}$ $(0<x<1)$	→	Bauxite
$Fe_2O_3.3H_2O$	→	Limonite
Cu_2O	→	Cuprite or Ruby copper
MnO_2	→	Pyrolusite
SnO_2	→	Tinstone or Cassiterite
TiO_2	→	Rutile
$Fe.Cr_2O_4$ $(FeO + Cr_2O_3)$	→	Chromite ore
$Na_2B_4O_7.10H_2O$	→	Borax or Tincal
$Ca_2B_6O_{11}.5H_2O$	→	Colemanite
U_3O_8	→	Pitch Blende
$FeO.TiO_2$	→	Ilmenite
ZnO	→	Zincite (Philosophers wool)
$3BeO.\ Al_2O_3.\ 6SiO_2$	→	Beryl

Oxy Salt Ore:

(i) Carbonate Ore:

$CaCO_3$	→	Lime stone
$MgCO_3$	→	Magnesite
$CaCO_3.MgCO_3$	→	Dolomite
$FeCO_3$	→	Siderite
$ZnCO_3$	→	Calamine
$Cu(OH)_2.CuCO_3$	→	Malachite or Basic Copper Carbonate
$Cu(OH)_2.2CuCO_3$	→	Azurite
$PbCO_3$	→	Cerrusite
$SrCO_3$	→	Strontianite

(ii) Sulphate ore:

$CaSO_4.2H_2O$	→	Gypsum
$MgSO_4.7H_2O$	→	Epsom Salt
$PbSO_4$	→	Anglesite
$BaSO_4$	→	Baryte
$Na_2SO_4.10H_2O$	→	Glauber Salt
$SrSO_4$	→	Celestine
$K_2SO_4.Al_2(SO_4)_3.24H_2O$	→	Potash Alum

(iii) Nitrate ore:

KNO_3	→	Indian Salt petre
$NaNO_3$	→	Chile Salt petre

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Note:

Kaolinite (a form of clay) $\rightarrow Al_2(OH)_4Si_2O_5$

Metallurgy:

The scientific and technological process used for the extraction/isolation of the metal from its ore is called as metallurgy.

Types of metallurgy:

(a) Pyro metallurgy :

Extraction of metal from ore by using heat energy.

Steps involved are **:** Calcination, roasting, reduction etc.

Ex. Less reactive metals : Cu, Fe, Co, Ni, Zn, Sn, Pb etc.

(b) Hydro metallurgy :

(Ag, Au, Cu)

Ag and Au ——→ By cyanide process.

Cu ——→ Pyro + Hydro

Steps are :

(i) Complex formation : $AgCl \xrightarrow{NaCN} \underset{\text{(sodium argento cynide)}}{Na[Ag(CN)_2]}$

(ii) Metal displacement : $2Na\,[Ag(CN)_2] \xrightarrow{Zn} Ag\downarrow + \underset{\text{(Impure)}}{Na_2[Zn(CN)_4]}$

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(c) Electrical metallurgy :

This process used for highly electro positive metal (s-block and Aℓ) obtained by electrolysis of fused salt/anhydrous medium.

Steps Involved In The Extraction Of Metals:

The isolation and extraction of metals from their ores involve the following major steps:

(A) Crushing of the ore.

(B) Dressing or concentration of the ore.

(C) Conversion of ore into their oxides

(D) Reduction of ore to the metal

(E) Purification or refining of the metal

(A) Crushing and Grinding :

The ore is first crushed by crushers and ground to a powder.

(B) Concentration of the ore:

Removal of the unwanted materials (e.g., sand, clays, etc.) from the ore is known as concentration, dressing or benefaction. Before proceeding for concentration, ores are graded and crushed to reasonable size. Concentration of ores involves several steps and selection of these steps depends upon the differences in physical properties of the compound of the metal present and that of the gangue. The type of the metal, the available facilities and the environmental factors are also taken into consideration. Some of the important procedures for concentration of ore are described below.

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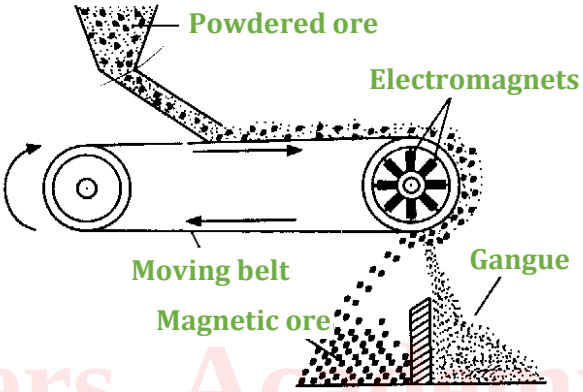
(1) By physical separation Method:

(i) Hydraulic washing or Gravity separation or Levigation method:

This is based on the difference between specific gravities of the ore and the gangue particles. It is therefore a type of gravity separation. In one such process, an upward stream of running water is used to wash the powdered ore. The lighter gangue particles are washed away and the heavier ore particles are left behind.

(ii) Electromagnetic separation :

This is based on differences in magnetic properties of the ore components. If either the ore or the gangue is attracted towards magnetic field, then the separation is carried out by this method. For example iron ores are attracted towards magnet, hence, non–magnetic impurities can be separated from them using magnetic separation. The powdered ore is dropped over a conveyer belt which moves over a magnetic roller Magnetic substance remains attracted towards the belt and falls close to it.



Electromagnetic separation

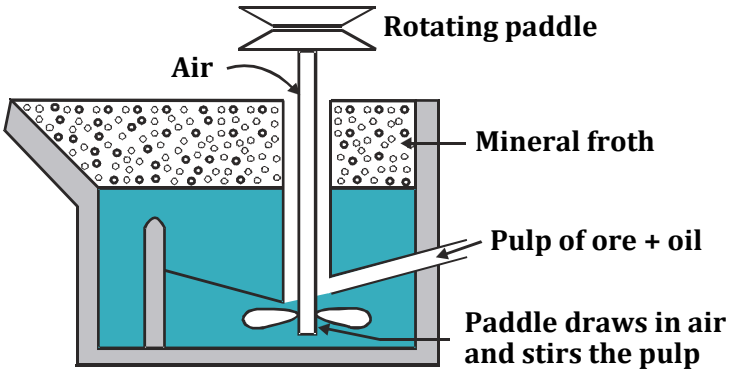
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Examples :

Chromite ore($FeO.Cr_2O_3$) is separated from non–magnetic silicious impurities and cassiterite ore(SnO_2) is separated from magnetic Wolframite ($FeWO_4 + MnWO_4$).

(iii) Froth floatation process :

This method is commonly used for the concentration of the low grade sulphide ores like galena, PbS; copper pyrites $Cu_2S.Fe_2S_3$ or $CuFeS_2$ (ore of copper) ; zinc blende, (ZnS) etc., and is based on the fact that gangue and ore particles have different degree of wettability with water and oil; the gangue particles are preferentially wetted by water while the ore particles are wetted by oil. In this process one or more chemical frothing agents are added.



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(a) Frothers :

These form stable froth which rises to the top of the floatation cell. Oil like pine oil, camphor oil etc., are used as frothers. These are added in small quantity.

(b) Froth stablizer : Aniline and Cresol :

The stabiliser are added to the frothers so that the froth can be stable for longer period.

(c) Collectors: Pine oil, Fatty acids, Potassium or sodium ethyl xanthate is used as a collector. These get attached with the particles of the sulphide ore and thus make them water-repellent. Consequently the ore particles pass on into the froth. Collectors are always added in small quantity.

(d) Activating and depressing agents :

When a mineral contains other minerals as impurities. The addition of these agents activates or depresses the floatation property of other minerals present as impurities and thus helps in separating the impurities. For example galena (PbS) usually contains the minerals namely zinc blende (ZnS) and pyrites (FeS_2) as impurities. Floatation is carried out by using potassium ethyl xanthate (used as a collector) along with NaCN and Na_2CO_3 (used as depressing agent).

The addition of NaCN and Na_2CO_3 depresses the floatation property of ZnS and FeS_2 grains, so mainly PbS passes into the froth when air is blown in. After PbS has been collected with the froth, the process is repeated by adding $CuSO_4$ (activator) which activates the floatation property of ZnS grains which are now removed with the froth. The acidification of the remaining material left in the floatation cell leads to the floatation of FeS_2.

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(2) By Chemical Separation Method:

Some of the ores are concentrated by means of chemical treatment.

Leaching :

It involves the treatment of the ore with a suitable reagent. Example: acids, bases and suitable chemical reagents. as to make it soluble while impurity remain insoluble. The ore is recovered from the solution by suitable chemical method.

(i) Bayer's process (Leaching of alumina from bauxite)

The principal ore of aluminium, bauxite, usually contains SiO_2, iron oxide and titanium oxide (TiO_2) as impurities. Concentration is carried out by digesting the powdered ore with a concentrated solution of NaOH at 473 - 523 K and 35 - 36 bar pressure. This way, Al_2O_3 is leached out as sodium aluminate (and also SiO_2 as sodium silicate) leaving behind the impurities, iron oxide and titanium oxide.

$$Al_2O_3(s) + 2NaOH(aq) + 3H_2O \longrightarrow 2Na[Al(OH)_4](aq)$$

The aluminate in solution is neutralised by passing CO_2 gas and hydrated Al_2O_3 is precipitated.

$$2Na[Al(OH)_4](aq) + CO_2(g) \longrightarrow Al_2O_3.xH_2O(s) + 2NaHCO_3(aq)$$

The sodium silicate remains in the solution and hydrated alumina is filtered, dried and heated to give back pure Al_2O_3 :

$$Al_2O_3.xH_2O(s) \xrightarrow[\text{heat}]{1470K} Al_2O_3(s) + xH_2O(g)$$

These steps comprise the Bayer's process.

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(ii) Cyanide process

In the metallurgy of silver and that of gold, the respective metal/ore is leached with a dilute solution of NaCN or KCN in the presence of air (or O_2) from which the metal is obtained later by displacement with zinc scrap.

$4M(s) + 8CN^{\ominus}(aq) + 2H_2O + O_2(g) \longrightarrow 4[M(CN)_2]^{\ominus}(aq)^{\oplus} + 4OH^{\ominus}(aq)$ (M= Ag or Au)

$2[M(CN)_2]^{\ominus}(aq) + Zn(s) \longrightarrow [Zn(CN)_4]^{2\ominus}(aq) + 2M(s)$

(C) Conversion of ore into their Oxides:

(1) Calcination

Calcination is a process in which ore is heated, generally in the **absence of air**, to expel water from a hydrated oxide or carbon dioxide from a carbonate at temperature below their melting points.

For Example :

(i) $Al_2O_3.2H_2O \xrightarrow{\Delta} Al_2O_3 + 2H_2O$, $2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O$

(ii) $2Fe_2O_3.3H_2O \xrightarrow{\Delta} 2Fe_2O_3 + 3H_2O$

(iii) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

(iv) $CaCO_3.MgCO_3 \xrightarrow{\Delta} CaO + MgO + 2CO_2\uparrow$

(v) $MgCO_3 \xrightarrow{\Delta} MgO + CO_2\uparrow$

(vi) $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2\uparrow$

(vii) $FeCO_3 \xrightarrow{\Delta} FeO + CO_2\uparrow$

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Advantages of Calcination :

(i) Moisture is removed.
(ii) Organic matter is destroyed
(iii) The hydroxide and carbonates ores are converted into their oxides.
(iv) The metal become porous and easily workable.

(2) Roasting

(Metal sulphides $\longrightarrow$ Metal oxide + SO_2)

The removal of the excess sulphur contained in sulphide ores by heating **in an excess of air** is called roasting.

The concentrated sulphide ore is heated in reverberatory furnace, below its melting point or fusion temperature in the presence of an excess of air with or without the addition of an external substance. In roasting definite chemical like oxidation, chlorination etc. take place but in calcination does not occur any major chemical changes.

(i) $2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$
(ii) $ZnS + 2O_2 \longrightarrow ZnSO_4$
(iii) $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + SO_2$
(iv) $4FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$
(v) $HgS + O_2 \longrightarrow HgO + SO_2$
(vi) $2As_2S_3 + 9O_2 \longrightarrow 2As_2O_3 + 6SO_2$

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Advantages of Roasting :

(i) Excess of sulphur is removed as volatile oxide.

$S + O_2 \rightarrow SO_2$ (air)

(ii) The metal sulphide is converted into metal oxide.

(iii) Impurities of arsenic and antimony are removed as their volatile oxides.

$Sb_4 + 3O_2 \rightarrow 2Sb_2O_3$

$As_4 + 3O_2 \rightarrow 2As_2O_3$

In PbS or ZnS

$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$

$PbS + 2O_2 \longrightarrow PbSO_4$

Thermodynamics of reduction processes (Ellingham diagram):

The extraction of metals from their oxides using carbon or other metals, and by thermal decomposition, involves a number of points which merit detailed discussion.

For a spontaneous reaction, the free energy change ΔG must be negative.

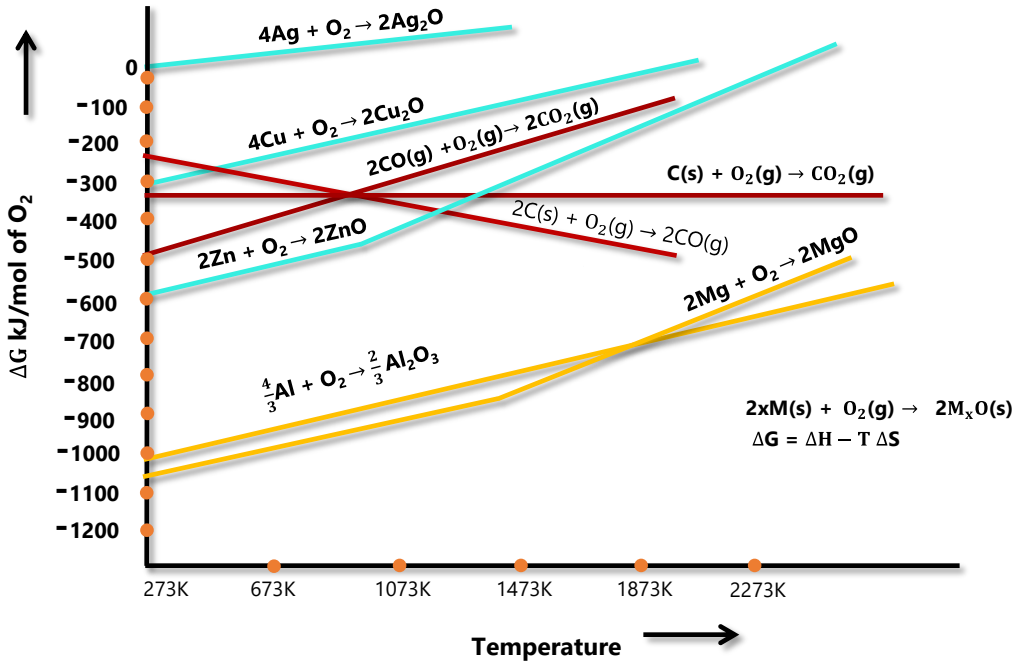
$\Delta G = \Delta H - T\Delta S$

ΔH is the enthalpy change during the reaction, T is the absolute temperature, and ΔS is the change in entropy during the reaction. Consider a reaction such as the formation of an oxide:

$M + O_2 \rightarrow MO$

Dioxygen is used up in the course of this reaction. Gases have a more random structure (less ordered) than liquids or solids. Consequently, gases have a higher entropy than liquids or solids. In this reaction S the entropy or randomness decreases, the hence ΔS is negative. Thus if the temperature is raised then $T\Delta S$ becomes more negative. Since $T\Delta S$ is subtracted in the equation, then ΔG becomes less negative. Thus the free energy changed increases with an increase of temperature.

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The free energy changes that occur when one gram molecule of a common reactant (in this case dioxygen) is used may be plotted graphically against temperature for a number of reactions of metals of their oxides. This graph is shown in figure and is called an Ellingham diagram (for oxides). Similar diagrams can be produced for one gram molecule of sulphur, giving an Ellingham diagram for sulphides, and similarly for halides.

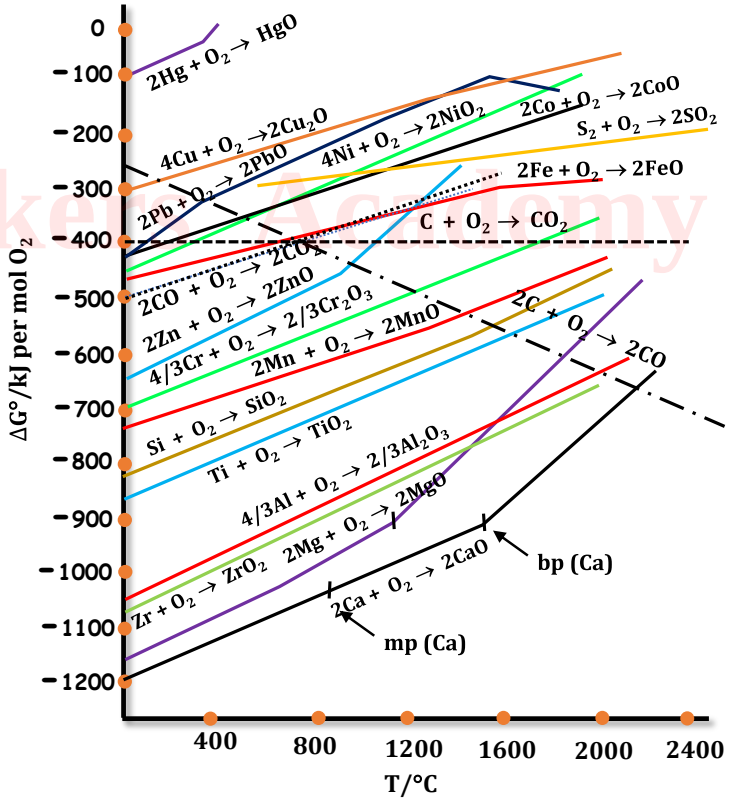
The Ellingham diagram for oxides shows several important features :

(i) The graph for metal oxide all slope upwards, because the free energy change increases with an increase of temperature as discussed above.

(ii) The free energy changes all follows a straight line unless the materials metal or vaporize.

(iii) When the temperature is raised, a point will be reached where the graph crosses the $\Delta G = 0$ line. Below this temperature the free energy of formation of the oxide is negative, so the oxide is stable. Above this temperature the free of formation of the oxide is positive, and the oxide becomes unstable, and should decompose into the metal and dioxygen.

(iv) Any metal will reduce the oxide of other metals which lie above it in the Ellingham diagram because the free energy will become more negative by an amount equal to the different between the two graphs at that particular temperature.



Limitations of Ellingham Diagram:

(i) The graph simply indicates whether a reaction is possible or not i.e., the tendency of reduction with a reducing agent is indicated. This is so because it is based only on the thermodynamic concepts. It does not say about the kinetics of the reduction process (Cannot answer questions like how fast it could be ?).

(ii) The interpretation of ΔG° is based on K($\Delta G^\circ = -RT \ln K$). Thus, it is presumed that the reactants of products are in equilibrium.

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Illustration 1:

Carbon monoxide is more effective reducing agent than carbon below 983 K but above this temperature the reverse is true. How would you explain this?

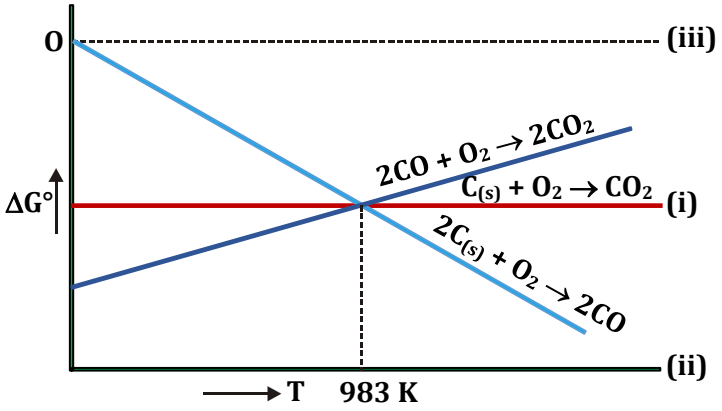
Solution:

The two reactions are as :

$C_{(s)} + O_2 \longrightarrow CO_2$ ...(i)

$2C_{(s)} + O_2 \longrightarrow 2CO$ ...(ii)

$2CO + O_2 \longrightarrow 2CO_2$ ...(iii)



Below 983 K reaction $2CO + O_2 \longrightarrow 2CO_2$ is more favoured due to more negative ΔG value thus CO is better reducing agent than carbon.

Above 983 K, reaction $2C_{(s)} + O_2 \longrightarrow 2CO$ has more negative ΔG than oxidation of CO to CO_2, so carbon will be better reducing agent.

Illustration 2:

Why Al can reduce oxide of Cr but not reduce oxide of Mg and Ca at 800°C ?

Solution:

A metal can reduce the oxide of metal placed above in the diagram. Due to more negative free energy. Such that, Al can reduce Cr but not Mg or Ca.

Illustration 3:

In what condition extraction of Mg is possible by MgO using carbon reduction method ?

Solution:

According to Ellingham diagram reduction of metal oxide with carbon will occur when the accompanying ΔG is negative. Such reduction become feasible at very high temperature. MgO can only be reduced by carbon at the temperature approximate 2000°C.

Mg form by heating MgO and C at 2000°C, at this temperature or above C reduces MgO. The gaseous mixture of Mg and CO was then cooled very rapidly to deposit the metal. This 'quenching' or 'shockcooling' was necessary as the reaction is reversible, and if cooled slowly the reaction will come to equilibrium further to the left and we get Mg.

$MgO + C \rightleftharpoons Mg + CO$

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(D) Reduction of ore to metal:

The calcined or roasted ore is then reduced to the metallic state in either of the following ways.

(1) Reduction by carbon (Smelting) :

(This is common method of reduction)

"Reduction of the oxide with carbon at high temperature is known as smelting".

The oxides of less electropositive metals like Pb, Zn, Fe, Sn, Cu etc. are reduced by strongly heating them with coal or coke, in the blast furnace.

Slag : Fusible material during reduction process.

Slag : Gangue + substance (to remove gangue)

Flux : Substance used for removing gangue

Acidic Flux: Borax, SiO_2 (remove basic impurity)

Basic Flux: MgO, $MgCO_3$, $CaCO_3$ (remove acidic impurity)

Smelting

- Concentrate ore $[\text{ore} + \text{gangue}] + \underset{(\text{R.A.})}{\text{carbon}} + \text{Flux} \rightarrow \text{Metal} + \text{Slag} + \text{Gases}$
- Cr_2O_3, Mn_3O_4, MnO_2 $\longrightarrow$ Carbon is not used for reduction
- Coke is not used for reduction of s-block oxide Al_2O_3 (due to formation of metal carbides)

  $CaO + 2C \longrightarrow CaC_2 + CO$

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Some reactions

(i) $CuO + CO \longrightarrow CO_2 + Cu$

$PbO + C \longrightarrow Pb + CO$

$Fe_2O_3 + 3C \longrightarrow Fe + 3CO$

$ZnO + C \longrightarrow Zn + CO$

$ZnO + CO \longrightarrow Zn + CO_2$

(ii) Gangue + acidic impurity (flux) $\rightarrow$ (slag)

$SiO_2 + CaCO_3 \longrightarrow CaSiO_3 + CO_2\uparrow$

$P_2O_5 + 3CaO \longrightarrow Ca_3(PO_4)_2$

(iii) Basic impurity + Flux $\longrightarrow$ slag

$MgCO_3 + SiO_2 \longrightarrow MgSiO_3 + CO_2\uparrow$

$FeO + SiO_2 \longrightarrow FeSiO_3$

Note:

Blue flame is obtained by burning of CO in smelting process.

(2) Self reduction : (For ores of Cu, Pb, Hg etc.)

Compounds of certain metals are reduced to metal without using any additional reducing agent.

Their sulphide ores are partially roasted to give some oxide. This oxide is now reduced to the metal by the remaining sulphide ore at elevated temperatures in the absence of air. The process is known as self reduction.

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Self reduction for Pb :

(i) $\underset{\text{(Galena)}}{2PbS} + \underset{\text{(air)}}{3O_2} \xrightarrow{\text{Roasting}} 2PbO + 2SO_2 \uparrow$

$\underset{\text{(unroasted ore)}}{PbS} + \underset{\text{(roasted ore)}}{2PbO} \xrightarrow[\text{Absence of air}]{\text{High temp}} 3Pb + \underset{\text{(Self reduction)}}{SO_2} \uparrow$

(ii) $2Cu_2O + Cu_2S \xrightarrow{\Delta} 6\,Cu + SO_2$

(3) Metal displacement method :

In this method, a water-soluble compound is obtained from the ore. The aqueous solution of the compound is reacted with a more electropositive metal which displaces, the metal from the solution.

(i) Zairvogel process for silver.

$\underset{\text{Argentite}}{Ag_2S} + 2O_2 \xrightarrow{85°C} Ag_2SO_4$

$Ag_2SO_4(aq) + \underset{\text{(Scrap Copper)}}{Cu} \longrightarrow CuSO_4(aq) + 2Ag(s)$

(ii) Seperation Ag by Complex formation (Cyanide process)

Silver and gold are extracted by a method involving complex formation.

$\underset{\text{(Powdered argentite)}}{Ag_2S} + 4NaCN \xrightarrow{\text{air}} 2Na[Ag(CN)_2] + Na_2SO_4$

$2Na[Ag(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4](aq) + \underset{\text{Black ppt.}}{2Ag\downarrow}$

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(4) Electrolytic reduction :

This process is mainly used for the extraction of **highly electropositive metals.**

Ex. Na, K, Mg, Ca, Al, etc.

Electrolysis is carried out in a large cells and a small amount of another suitable electrolyte is added which:

(i) Lowers the melting point of the main electrolyte

(ii) Enhances its conductivity

(iii) Reduces corrosion troubles

e.g. Manufacture of metallic sodium (Down's process)

Molten NaCl containing a little $CaCl_2$ is electrolyzed between graphite anode and iron cathode. The various reactions that take place are

On Fusion : $NaCl \rightleftharpoons Na^{\oplus} + Cl^{\ominus}$ (Ions become mobile)

On Electrolysis : **At Cathode :** $Na^{\oplus} + e^{\ominus} \rightarrow \underset{\text{(Metallic sodium)}}{Na}$

At Anode : $2Cl^{\ominus} \rightarrow Cl_2(g) + 2e^{\ominus}$

(5) Reduction by Al (Thermite Reduction) :

This process is employed in the case of those metals which have very high melting points and are to be extracted from their oxides

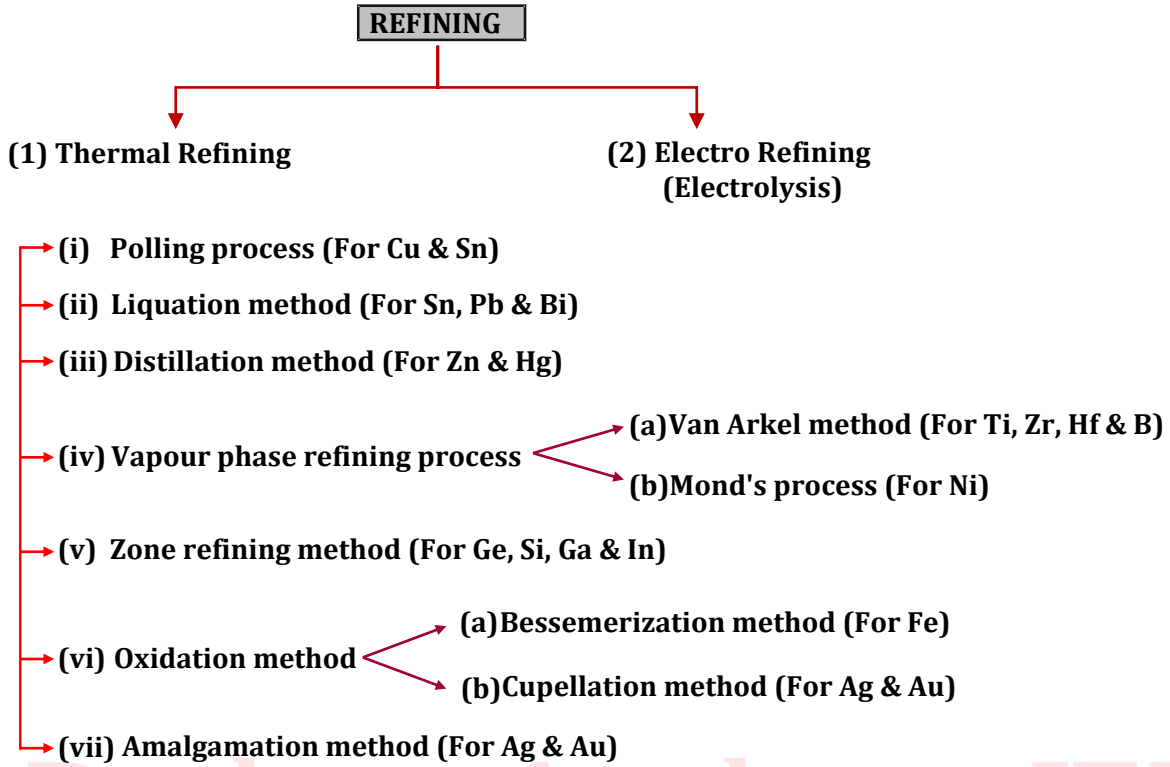
$Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$

$3Mn_3O_4 + 8Al \longrightarrow 9Mn + 4Al_2O_3$

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(E) Refining of Metals:

Metals obtained by the reduction of its compound still contains some objectionable substance and have to be refined.



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Depending upon the nature of the metal and impurities, the following methods are used for purification of the metals.

(1) Thermal Refining:

(i) Poling process : This process is used for the purification of copper and tin.

(a) Purification of impure copper :

Impure copper is remelted in a reverberatory furnace lined with SiO_2 and a blast of O_2 is blows into the furnace. O_2 oxidises S, Sb and As to their respective oxides which, being, volatile, get volatilised and are thus removed. Fe is oxidised to FeO which forms a slag of $FeSiO_3$ with SiO_2 lining of the furnace. Molten copper left behind contains CuO as impurity. This molten copper is treated with powdered anthracite and then stirred with a pole of green wood. Green wood, at high temperature, liberates hydrocarbon gases, which are converted into methane (CH_4). Methane thus obtained reduces CuO to free Cu–metal, which is about 99.5% pure and is called **tough pitch copper**.

Green wood $\rightarrow$ Hydrocarbons (CH_4)

$$4CuO + CH_4 \rightarrow \underset{\text{(pure metal)}}{4Cu} + CO_2 + 2H_2O$$

(b) Purification of impure tin :

Impure tin metal contains the impurities of Cu, Fe, W and SnO_2. The impurity of SnO_2 is due to the incomplete reduction of tin stone ore (SnO_2) during smelting. In order to remove these impurities, the impure molten tin metal is taken in a big pot and stirred with a pole of green wood. Green wood, at high temperature liberates hydrocarbon gases, which are converted into methane CH_4. Methane thus obtained reduces SnO_2 to pure metal while the impurities of Cu, Fe, W etc. come up to the

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surface, where they come in contact with air and are oxidised to their respective oxides. The oxides form a scum on the surface of pure tin metal. This scum is removed from the surface. Tin metal obtained by this method is 99% pure.

Green wood $\rightarrow$ Hydrocarbon (CH_4)

$2SnO_2 + CH_4 \rightarrow 2Sn + CO_2 + 2H_2O$

(ii) Liquation:

This method is used for the refining of metals having low melting point and are associated with high melting impurities.

Ex. Pb, Sn, Sb, Bi and Hg.

The impure metal is heated on the sloping hearth of a furnace.

The pure metal flows down leaving behind the non- fusible material on the hearth.

(iii) Distillation:

Metals having low boiling point are refined by this method.

Example: zinc, cadmium and mercury.

(iv) Vapour phase refining:

In this method, the metal is converted into its volatile compound which is collected and decomposed to give pure metal. So, the two requirements are:

(i) the metal should form a volatile compound with an available reagent,

(ii) the volatile compound should be easily decomposable, so that the recovery is easy.

Following examples will illustrate this technique.

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(a) Mond Process for Refining Nickel: In this process, nickel is heated in a stream of carbon monoxide forming a volatile complex named as nickel tetracarbonyl. This complex is decomposed at higher temperature to obtain pure metal.

$Ni + 4CO \xrightarrow{330-350\ K} Ni(CO)_4$

$Ni(CO)_4 \xrightarrow{450-470\ K} Ni + 4CO$

(b) Van Arkel Method for Refining Zirconium or Titanium: This method is very useful for removing all the oxygen and nitrogen present in the form of impurity in certain metals like Zr and Ti. The crude metal is heated in an evacuated vessel with iodine. The metal iodide being more covalent, volatilises: The metal iodide is decomposed on a tungsten filament, electrically heated to about 1800K. The pure metal deposits on the filament.

$Zr + 2I_2 \rightarrow ZrI_4$

$ZrI_4 \rightarrow Zr + 2I_2$

For Titanium, the reaction is

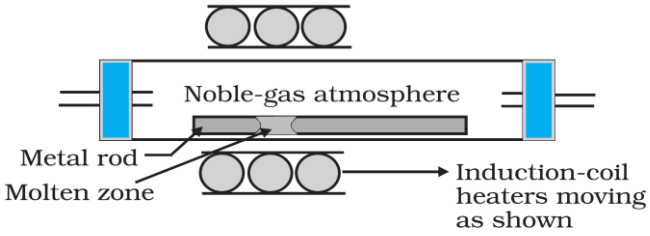
$\underset{\text{Impure}}{Ti(s)} + 2I_2\,(g) \xrightarrow{523\ K} Ti\,I_4(g)$

$TiI_4(g) \xrightarrow{1700\ K} Ti(s) + 2I_2(g)$

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(v) Zone refining or fractional crystallisation:

This method is based on the principle that the impurities are more soluble in the melt than in the solid state of the metal. A mobile heater surrounding the rod of impure metal is fixed at its one end. The molten zone moves along with the heater which is moved forward. As the heater moves forward, the pure metal crystallises out of the melt left behind and the impurities pass on into the adjacent new molten zone created by movement of heaters. The process is repeated several times and the heater is moved in the same direction again and again. Impurities get concentrated at one end. This end is cut off. This method is very useful for producing semiconductor and other metals of very high purity, e.g., germanium, silicon, boron, gallium and indium.



Zone refining process

(vi) Oxidation Process (Pyrometallurgical oxidation process):

These processes are used for refining of metals associated with impurities having high affinity for oxygen than the metal itself.

Cupellation, puddling and bessemerization are important oxidation processes employed for refining different metals.

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(a) Bessemerisation (Purification of iron):

The iron obtained from a blast furnace is a brittle material called **cast iron** or **pig iron**. It contains about 4% elemental C and smaller amounts of other impurities such as elemental Si, P, S, and Mn that are formed from their compounds in the reducing atmosphere of the furnace. The most important of several methods for purifying the iron and converting it to steel is the basic oxygen process or oxidative refining. Molten iron from blast furnace is exposed to a jet of pure O_2 gas for about 20 minutes in a furnace that is lined with basic oxide such as CaO. The impurities in the iron are oxidized and the acidic oxides that form react with basic CaO to yield a molten slag that can be poured off. Phosphorous, for example, is oxidized to P_4O_{10}, which then reacts with CaO to give molten $Ca_3(PO_4)_2$.

$$P_4 + 5\,O_2 \longrightarrow P_4O_{10}$$

$$\underset{\text{Basic oxide}}{6\,CaO} + \underset{\text{acidic oxide}}{P_4O_{10}} \longrightarrow \underset{\text{slag}}{2\,Ca_3(PO_4)_2}$$

Mn also passes into the slag because its oxide is basic and reacts with SiO_2 yielding molten manganese silicate.

This process produces steel that contains about 1% carbon but only very small amount of P and S. Usually the composition of liquid steel is monitored by chemical analysis and the amount of oxygen and impure iron used are adjusted to achieve the desired concentration of carbon and other impurities.

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(b) Cupellation (removal of lead):

In this process the molten impure metal is heated in a cupel, which is boat-shaped dish made of bone ash or cement, and a blast of air is passed over the molten metal. The impurities are oxidized and the volatile oxides thus produced escape with the blast of air. The pure metal remains behind in the cupel. Pb present in silver is removed by cupellation process.

$$2\,Pb(g) + O_2 \longrightarrow 2\,PbO(g)$$

(vii) Amalgamation process:

For nobel metal Au, Ag from the native ore.

ore powder + Hg $\longrightarrow$ Amalgam $\xrightarrow{\text{distilled}}$ Metal + Hg (vapour)

Some other methods:

(a) Kroll collen process

$$TiCl_4 + 2Mg \xrightarrow{\Delta} 2MgCl_2 + Ti$$

(b) IMI method

$$TiCl_4 + 4Na \xrightarrow{\Delta} 4NaCl + Ti$$

(2) Electro–refining of Metals:

Metals such as Cu, Ag, Zn, Sn, Pb, Al, Ni, Cr are refined by this method.

The impure metal is made the anode of a electrolytic cell, while cathode is thin plate of pure metal.

Electrolyte is the solution of a double salt of the metal.

On passing the electric current pure metal from the anode dissolves and gets deposited at the cathode.

The soluble impurities go into the solution while insoluble or less electropositive impurities settle down below the anode as **anode mud** or sludge.

Example: Impurities from blister coppor deposit as anode mud which contains Au, Ag, Sb, Te, Se and Pt.

(i) Electrorefining of Copper

Anode	:	Blister copper (98%)
Cathode	:	Pure copper
Electrolyte	:	An aqueous solution of $CuSO_4$ (15%) + 5% dil H_2SO_4

(ii) Electrorefining of Pb (Bett's process)

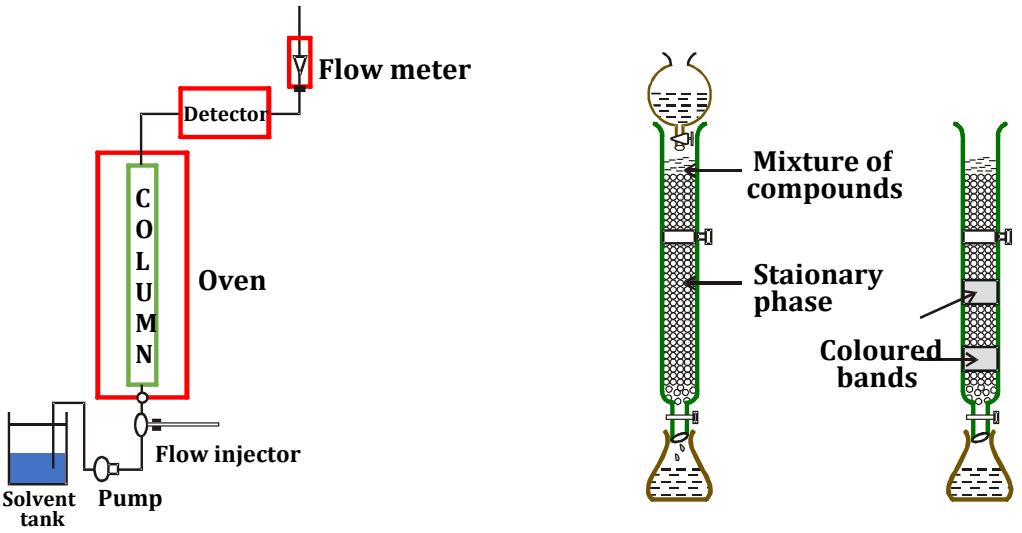
Anode	:	Impure lead
Cathode	:	Pure lead
Electrolyte	:	A mixture of $PbSiF_6$ and H_2SiF_6

(iii) Other method : Chromatographic method

This method is based on the principle that different components of a mixture are differently adsorbed on an adsorbent. The mixture is put in a liquid or gaseous medium which is moved through the adsorbent. Different components are adsorbed at different levels on the column. Later the adsorbed components are removed (eluted) by using suitable solvent (eluant). Depending upon the physical state of the moving medium and the adsorbent material and also on the process of passage of the moving medium, the chromatographic method is given the name. In one such method the column of Al_2O_3 is prepared in a glass tube and the moving medium containing a solution of the components is in liquid form.

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This is an example of column chromatography. This is very useful for purification of the elements which are available in minute quantities and the impurities are not very different in chemical properties from the element to be purified. There are several chromatographic techniques such as paper chromatography, column chromatography, gas chromatography, etc. Procedures followed in column chromatography have been depicted in the following figures.



(a) Industrial method **(b) Laboratory method**

Schematic diagrams showing column chromatography

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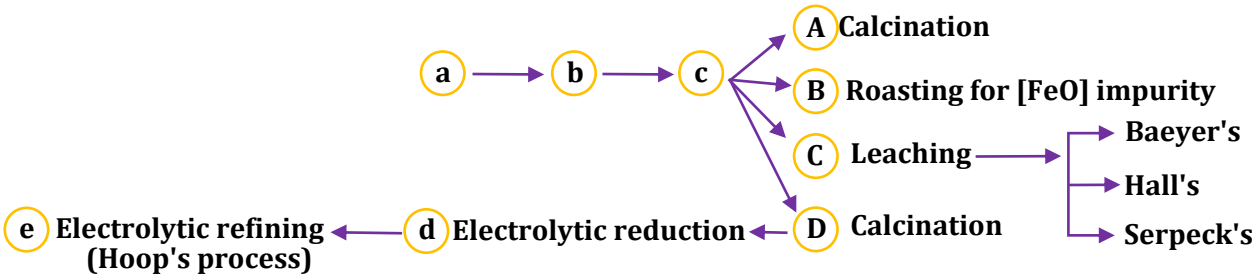
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EXTRACTION OF SOME INDIVIDUAL METALS

Extraction of Aluminium:

Ore - Bauxite $AlO_x(OH)_{3-2x}$ (where $0 < x < 1$)

Extraction of Aluminium:
Short chart of Al from $Al_2O_3.2H_2O$ (Bauxite)



Step-1: Concentration of Bauxite ore:

(a) Baeyer's process : (Used for red bauxite in which main impurity is iron oxide)

Bauxite is the principal ore of aluminium. It usually contains SiO_2, iron oxides and titanium oxide (TiO_2) as impurities. Concentration is carried out by heating the powdered ore with a concentrated solution of NaOH at 473 – 523 K and 35 – 36 bar pressure. This process is called **digestion**. This way, Al_2O_3 is extracted out as sodium aluminate. The impurity, SiO_2 too dissolves forming sodium silicate. Other impurities are left behind.

$Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \rightarrow 2Na[Al(OH)_4](aq)$

The sodium aluminate present in solution is neutralised by passing CO_2 gas and hydrated Al_2O_3 is precipitated. At this stage, small amount of freshly prepared sample of hydrated Al_2O_3 is added to the solution. This is called **seeding**. It induces the precipitation.

$2Na[Al(OH)_4](aq) + 2CO_2(g) \rightarrow Al_2O_3.xH_2O(s) + 2NaHCO_3\ (aq)$

Sodium silicate remains in the solution and hydrated alumina is filtered, dried and heated to give back pure Al_2O_3.

$Al_2O_3.xH_2O(s) \xrightarrow{1470\ K} Al_2O_3(s) + xH_2O(g)$

(b) Hall's Process : (Red bauxite)

Bauxite ore + $Na_2CO_3 \xrightarrow{\text{Fused}} NaAlO_2$

↓ extracted with water

Solution

↓ warmed 50° – 60° C

CO_2 is circulated

↓

$Al(OH)_3 + Na_2CO_3$

(c) Serpeck's process : (Used for white bauxite in which main impurity is silica)

Bauxite ore + coke + (Nitrogen) $N_2 \xrightarrow{1800°C} AlN \xrightarrow{H_2O} Al(OH)_3 + \underset{ppt}{NH_3}$

$C + SiO_2$ (gangue) $\rightarrow CO_2 + Si$

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Step-2 : Calcination:

$Al(OH)_3 \xrightarrow{1500°C} Al_2O_3$

Step-3 : Electrolytic Reduction:

Electrolyte Al_2O_3 dissolved in Na_3AlF_6 and CaF_2

Cathode–Carbon lining

Anode – Graphite rods

Electrolytic reduction (Hall-Heroult process) :

The purified Al_2O_3 is mixed with Na_3AlF_6 (cryolite) or CaF_2 (fluorspar)) which lowers the melting point of the mixture and brings conductivity. The fused matrix is electrolysed. Steel cathode and graphite anode are used. The graphite anode is useful here for reduction to the metal. The overall reaction may be taken as :

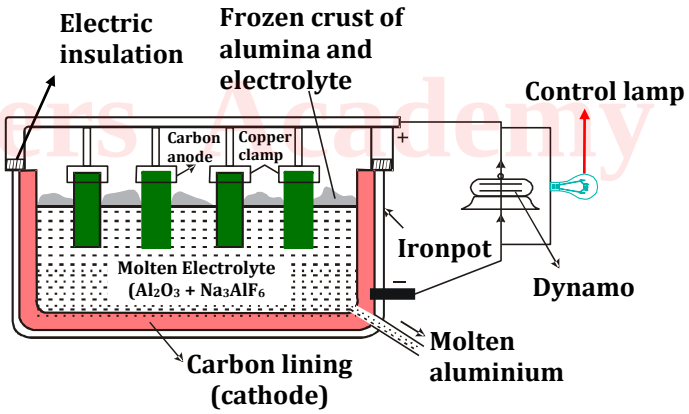
$$2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$$

The electolysis of the molten mass is carried out in an electrolytic cell using carbon electrodes. The oxygen liberated at anode reacts with the carbon of anode producing CO and CO_2. This way for each kg of aluminium produced, about 0.5 kg of carbon anode is burnt away. The electrolytic reactions are :

Cathode : $Al^{3\oplus}$ (melt) + $3e^{\ominus} \rightarrow Al(l)$

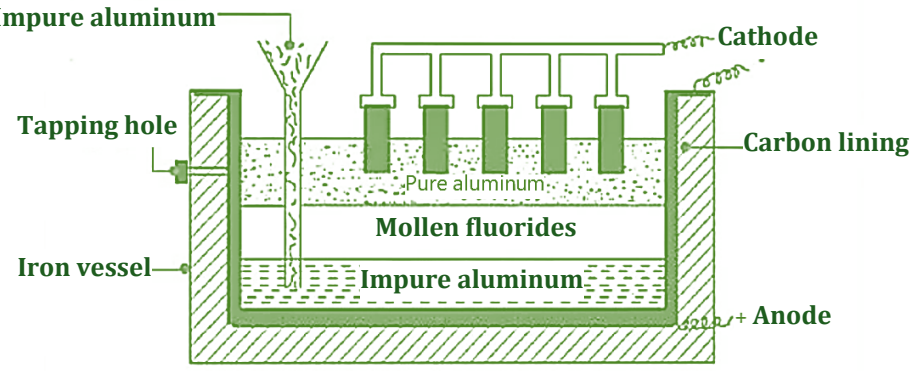
Anode : $C(s) + O^{2\ominus}$ (melt) $\rightarrow CO(g) + 2e^{\ominus}$

$C(s) + 2O^{2\ominus}$ (melt) $\rightarrow CO_2(g) + 4e^{\ominus}$



$Al_2O_3 \xrightarrow[950°C]{\text{Electrolysis}} Al + O_2$
$\downarrow$
99.8% pure

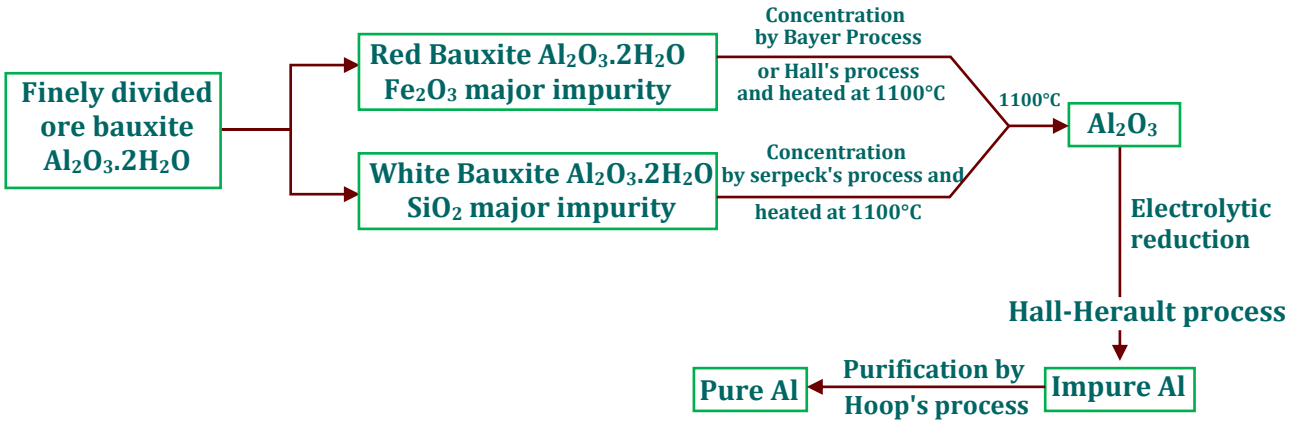
Step-4: Electrolytic Refining



(Hoop's process)
Pure Al (99.98 % pure)

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Flow chart of Al from $Al_2O_3.2H_2O$ (Bauxite)



Important points:

(a) Useful gas NH_3 is evolved in the leaching of bauxite by Serpeck's process.

(b) In the electrolytic reduction of Al_2O_3, cryolite (Na_3AlF_6) is added along with fluorspar (CaF_2) to
- decrease M.P. of Al_2O_3
- decrease viscosity of electrolyte (CaF_2 is used)
- increase conductivity

(c) In the electrolytic reduction graphite anode get corrode or finish due to reaction with O_2 liberates at anode, hence it had to be changed periodically.

(d) In the electrolytic refining (4th step) no electrodes are used. In the Hoop's process molten pure Al is used as cathode and molten impure Al is used as anode.

(e) In the Hoop's process carbon dust is sprayed over molten Al to
- avoid heat loss
- minimise metallic lustre (glaze) which is harmful for the eyes.

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Uses:

(i) Aluminium foils are used as wrappers for chocolates.

(ii) The fine dust of the metal is used in paints and lacquers.

(iii) Aluminium, being highly reactive, is also used in the extraction of chromium and manganese from their oxides.

(iv) Wires of aluminium are used as electricity conductors.

(v) Alloys containing aluminium, being light, are very useful.

Extraction of Zinc:

Occurrence:

Its important minerals are :-

(i) Zinc blende or black jack or Sphalerite	ZnS
(ii) Zincite	ZnO
(iii) Calamine	$ZnCO_3$

Extraction:

(i) Concentration :

The powdered ore ZnS is concentrated by froth floatation method.

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(ii) (a) Roasting:

Concentrated ZnS is roasted at 700°C. A mixture of ZnO and $ZnSO_4$ is obtained

$$ZnS \xrightarrow{\text{concentration}} ZnS \xrightarrow{\text{Roasting}} \begin{matrix} ZnO + SO_2 \uparrow \\ ZnSO_4 \end{matrix}$$

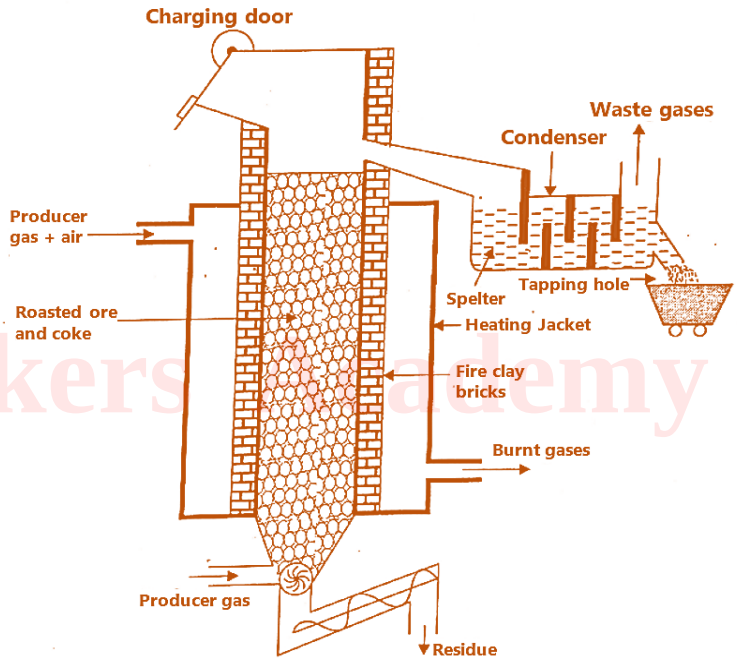
(b) Calcination Process

$$ZnCO_3 \xrightarrow{\text{concentration}} ZnCO_3 \xrightarrow{\text{calcination}} ZnO + CO_2$$

(iii) Carbon Reduction (Extraction of zinc from zinc oxide):

The reduction of zinc oxide is done using coke. The temperature in this case is higher than that in case of copper. For the purpose of heating, the oxide is made into brickettes with coke and clay.

$$ZnO + C\ (coke) \xrightarrow{1673K} Zn + CO$$



Vertical retort process for the reduction of ZnO to Zn – metal.

The metal is distilled off and collected by rapid chilling is called **zinc spelter**.

(iv) Electrolysis (Electrolytic refining)**:**

Anode → Impure Zn

Cathode → **Thin Al-rod**

Electrolyte → Solution of $ZnSO_4 + H_2SO_4$ (dil.)

on the electrolysis zinc is deposited at cathode. The metal is scrapped off and melted to give 99.95% pure metal.

Uses:

(i) In making alloys e.g. brass, German silver, electron (Alloy of Mg with smaller amount of Al, Y, Ag, Gd, Zn) etc.

(ii) In the extraction of silver and gold by cyanide process.

(iii) It is also used in large quantities in batteries and dry cells for making cathode container.

(iv) Zn–Cu couple, Zn- Hg, zinc dust etc. are used as reducing agent in organic reactions.

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(v) large amount of zinc is used for galvanizing iron. Zinc is deposited on the surface of iron articles. This process is called galvanization.

(vi) It is also used in large quantities in batteries, as a constituent of many alloys, e.g., brass, (Cu 60%, Zn 40%) and German silver (Cu 25-30%, Zn 25-30%, Ni 40–50%).

(vii) Zinc dust is used as a reducing agent in the manufacture of dye-stuffs, paints, etc.

Extraction of Copper:

Occurrence:

Copper occurs in free as well as in combined state. The main ores are as follows.

(i)	Copper pyrites	$CuFeS_2$ or $Cu_2S.\ Fe_2S_3$
(ii)	Cuprite (Ruby copper)	Cu_2O
(iii)	Copper glance	Cu_2S
(iv)	Malachite	$Cu(OH)_2.\ CuCO_3$
(v)	Azurite	$2Cu(OH)_2.\ CuCO_3$

Extraction:

Copper is extracted from the ores (sulphide) by two process.

(A) Pyrometallurgical Process :

This is a dry process and applied to high grade ores (containing 4% or more copper).

(B) Hydrometallurgical Process :

This is a wet process and applied to low grade ores.

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(A) Extraction from pyrites by pyrometallurgical process (Smelting Process)

(i) Concentration :

The finely powdered ore is concentrated by froth floatation process.

(ii) Roasting :

The concentrated ore is heated strongly in presence of air.

Sulphur, arsenic, and antimony are removed in form of their volatile oxides while the ore is converted into a mixture of cuprous and ferrous sulphides.

These sulphides are partially oxidised to oxides.

(a) $S + O_2 \rightarrow SO_2 \uparrow$

(b) $2As_2S_3 + 9O_2 \rightarrow 2As_2O_3\uparrow + 6SO_2 \uparrow$

(c) $2Sb_2S_3 + 9O_2 \rightarrow 2Sb_2O_3 \uparrow + 6SO_2 \uparrow$

(d) $2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$

$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$

(iii) Smelting

The roasted ore is mixed with sand (flux) and coke (fuel) and then heated in a reverberatory furnace water jacketed blast furnace called smelter. The oxidation of ferrous sulphide which started during roasting now goes a step further. Ferrous oxide formed, reacts with sand to form ferrous silicate (slag).

Here some FeS reacts with Cu_2O to form Cu_2S again.

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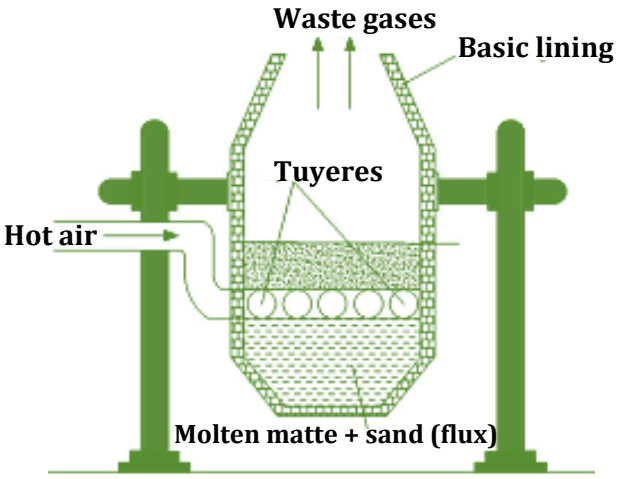
(a) $FeO + SiO_2 \rightarrow FeSiO_3$ (Slag)

(b) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$

(c) $Cu_2O + FeS \rightarrow Cu_2S + FeO$

Slag (forming upper layer) and molten mass consisting of Cu_2S and a little FeS **(forming lower layer and commonly called matte)** are removed from separate holes.

(iv) Bessemerisation :



Molten matte is heated in a Bessemer converter and a blast of air mixed with sand is blown through the molten mass.

Here iron (FeS) is completely removed as slag, a part of Cu_2S is oxidised to Cu_2O and a part of Cu_2S reduces Cu_2O to metallic copper.

(a) $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2 \uparrow$

$FeO + SiO_2 \rightarrow FeSiO_3$

(b) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2 \uparrow$

$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2 \uparrow$

The molten copper is poured off into moulds. As it cools it gives up the dissolved sulphur dioxide which forms blister on the surface of the metal. Hence the metal thus obtained is called blister copper and contains 98 percent copper.

(v) Refining of copper :

(a) Poling :

The molten blister copper is heated in pressure of air and stirred with green wood **poles**.

S and As are oxidised to their volatile oxides.

Iron is oxidised and forms a scum or slag which is **skimmed** off.

The cuprous oxide is reduced to Cu by CO and CH_4 from green wood.

$Cu_2O + CO \rightarrow 2Cu + CO_2$ $\qquad$ $4Cu_2O + CH_4 \rightarrow 8Cu + 2H_2O + CO_2$

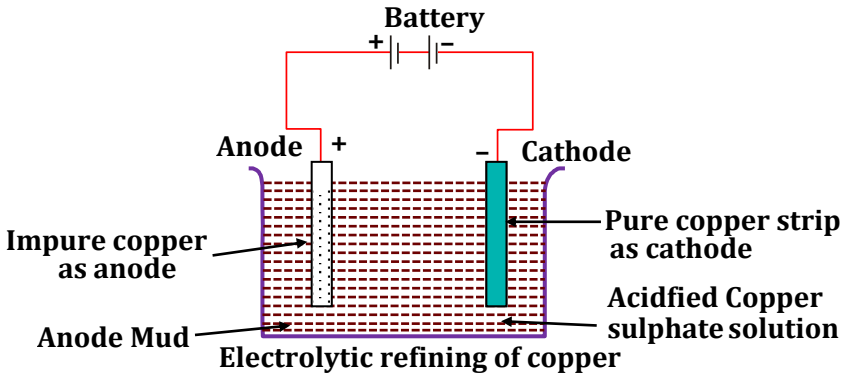
(b) Electrolytic refining :

The electrolytic bath contains an acidified solution of copper sulphate. Impure copper acts as anode while cathode is of the pure copper strip. When electric current is passed, there is transfer of pure copper from anode to cathode which gradually grows in size.

Impurities from the blister copper deposit as anode mud which contains Au, Ag, Sb, Te, Se & Pt.

The cathode is removed and copper of about 99.99% purity is obtained.

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(B) Hydrometallurgical Process:

Crushed and powdered low grade ore is exposed to air and water then copper sulphide is oxidised to copper sulphate. Some iron sulphate and sulphuric acid is also produced. The process is completed in about a year.

(a) $\underset{\text{(Copper pyrite)}}{CuFeS_2} + 4O_2 \rightarrow CuSO_4 + FeSO_4$

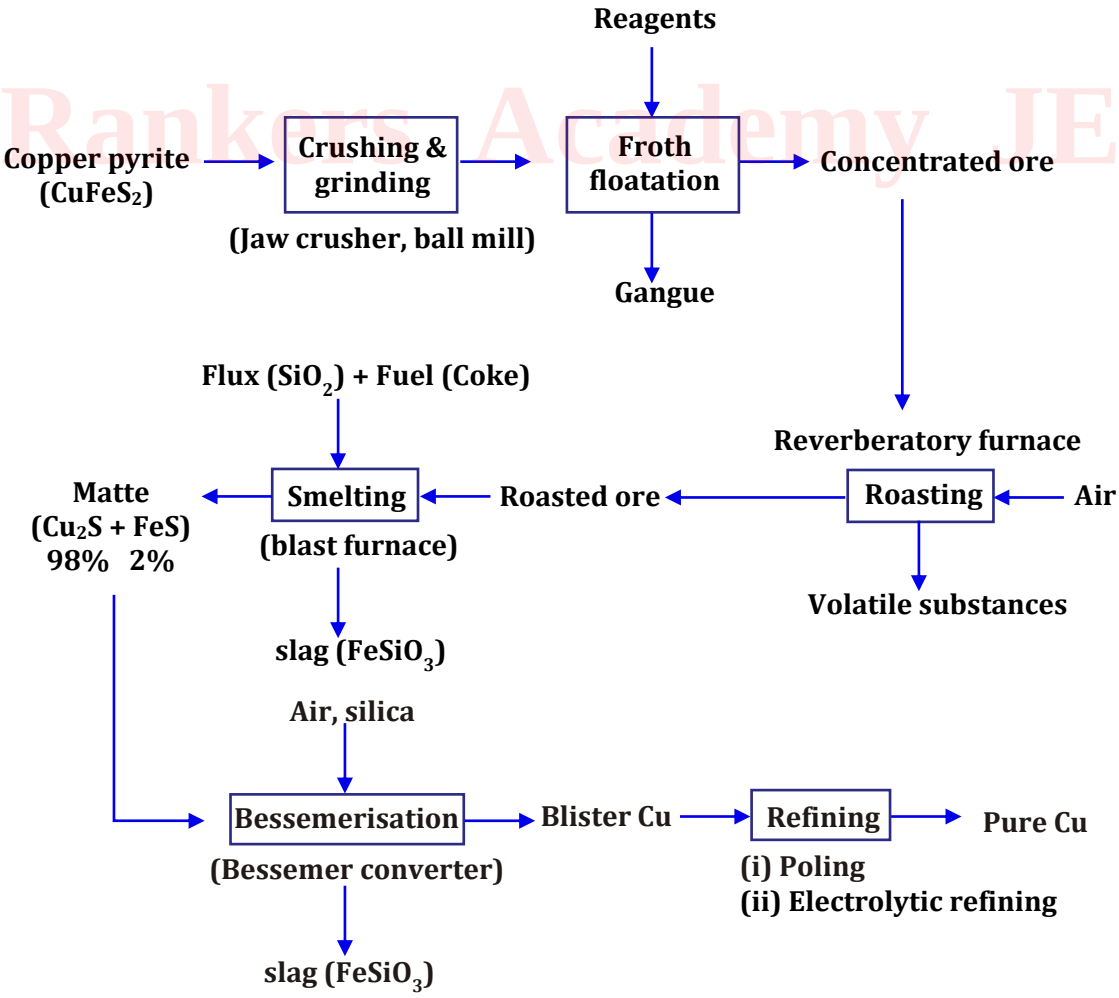
(b) $\underset{\text{(Copper glance)}}{2Cu_2S} + 5O_2 \rightarrow 2CuSO_4 + 2CuO$

(c) $\underset{\text{(malachite)}}{CuCO_3.Cu(OH)_2} + 2H_2SO_4 \rightarrow 2CuSO_4 + CO_2 + 3H_2O$

The pale green liquor draining from the bottom of the heaps is $CuSO_4$ which is collected in pans and treated with iron scraps which precipitate out copper.

$CuSO_4 + Fe/Zn/H_2 \rightarrow FeSO_4/ZnSO_4/H_2SO_4 + Cu$ (red ppt.)

Process in brief



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Uses :

Copper is the second most useful metal (the first being Iron) because of its stability in air and water and excellent conductivity.

It is used :–

(i) In the manufacture of electrical wires, cables etc.

(ii) For electroplating.

(iii) As a coinage metal and in ornaments and jewellery.

(iv) For the manufacture of alloys like brass (Cu + Zn), bronze (Cu + Sn) German silver (Cu + Zn + Ni) bell metal (Cu + Sn) gun metal (Cu + Sn + Zn), copper coins (Cu + Zn + Sn) etc.

Extraction of Iron:

Main Ore : Haematite (Fe_2O_3)

Some following steps are involved :

(i) Crushing & Grinding :

(ii) Concentration : By gravity separation method.

(iii) Roasting :

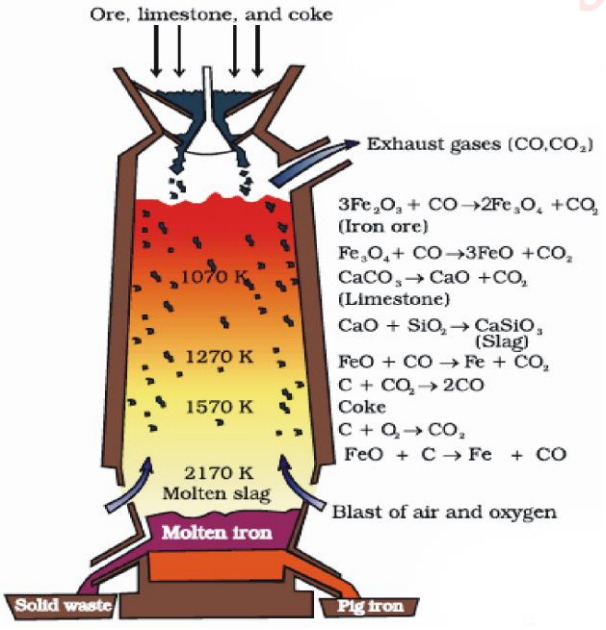
(iv) Carbon reduction or Smelting (Blast furnace) :

Pig iron is obtained from this process

The calcined ore (8 parts) mixed with coke, a reducing agent (4 parts) and lime stone a flux (1 part) is reduced in a blast furnace.

Since the blast furnace has different temperatures at different zones, different reactions take place at different zones.

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(a) Zone of combustion (1500 -1600°C)

This zone is near at the bottom of the furnace and little above the tuyers. It increases temperature of the furnace because of exothermic reactions.

$2C + O_2 \rightarrow 2CO$ $\quad\quad$ $C + O_2 \rightarrow CO_2$

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(b) Zone of fusion (1200- 1300°C)

It is just above the zone of combustion. Here the iron melts and trickles down in the hearth while the slag being lighter floats over the molten metal and thus prevents oxidation of Fe by blast of air.

(c) Zone of heat absorption or slag formation (800- 1000°C)

This is the middle part of the furnace. Here rising CO_2 is reduced to carbon monoxide.

$$CO_2 + C \rightarrow 2CO$$

The reaction being endothermic, lowers the temperature of the zone.

Limestone decomposes forming CaO which reacts with SiO_2 forming slag.

$$CaCO_3 \rightarrow CaO + CO_2 \qquad CaO + SiO_2 \rightarrow CaSiO_3 \text{ (Slag)}$$

(d) Zone of reduction (400-700°C)

It is near the top of the furnace. Here the calcined ore is reduced to Fe by rising CO.

$$Fe_2O_3 + CO \rightleftharpoons 2FeO + CO_2 \uparrow$$

$$Fe_3O_4 + CO \rightleftharpoons 3FeO + CO_2 \uparrow$$

$$FeO + CO \rightarrow Fe + CO_2 \uparrow$$

To retard the backward reaction, supply of excess of CO is maintained by the following reaction.

$$CO_2 + C \rightarrow 2CO$$

Since the temperature of this zone is too low to melt iron, the metal produced is known as spongy iron.

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(v) Refining :

Purification of Fe can be done by different method which are as follows :

(a) Puddling Process

(b) Bessemerisation Process

(c) Open hearth Process

(d) L. D. Process

Thus we got pure iron.

Types of Iron:

Pig Iron :

The iron obtained from Blast furnace contains about 4% carbon and many impurities in smaller amount (e.g., S, P, Si, Mn). This is known as pig iron. It can be moulded into variety of shapes.

Cast Iron :

Cast iron is different from pig iron and is made by melting pig iron with scrap iron and coke using hot air blast. It has slightly lower carbon content (about 3%) and is extremely hard and brittle.

Wrought Iron :

Wrought Iron or malleable iron is the purest form of commercial iron and is prepared from cast iron by oxidising impurities in a reverberatory furnace lined with haematite.

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Steel :

It is the most important form of iron and finds extensive applications. As far as carbon content (impurity) is concerned it is mid-way between cast iron and wrought iron, it contains (0.25- 2%) carbon.

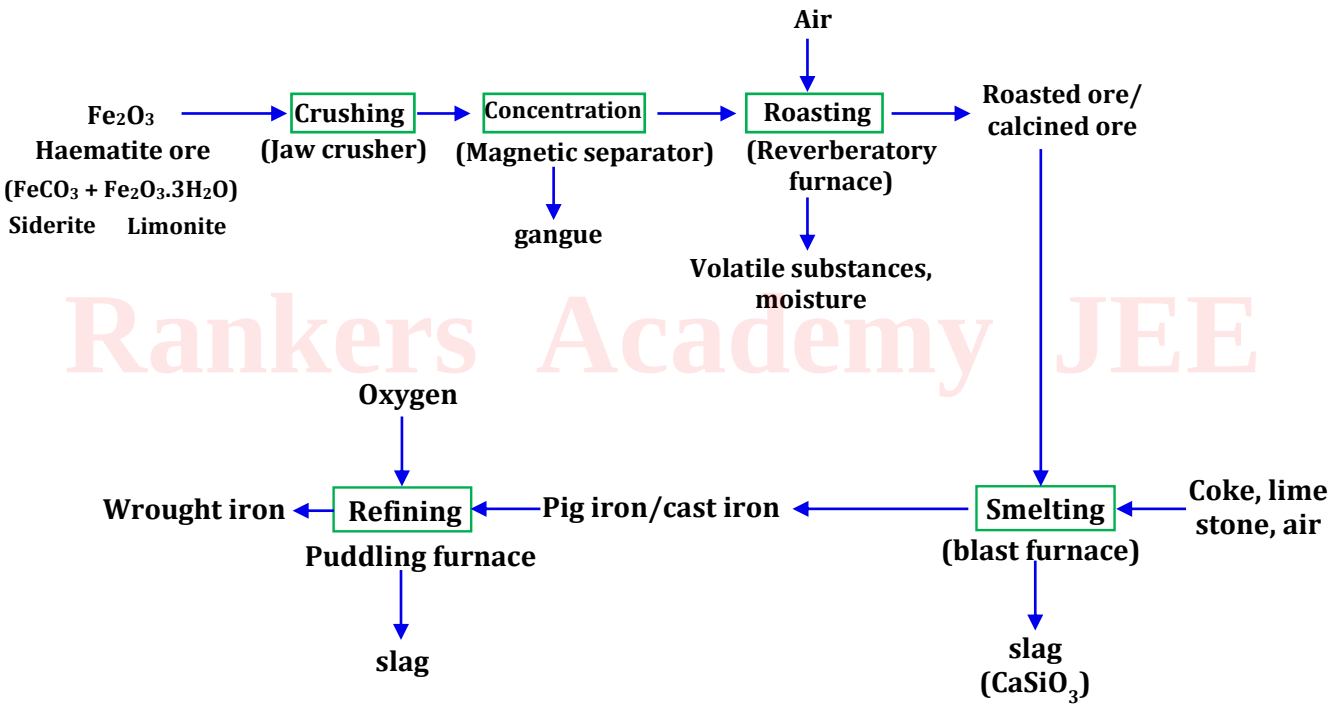
Thus, all the three forms of iron differ in their carbon contents, both iron and steel are obtained from cast iron.

Order of Melting Point : Wrought Iron > Steel > Cast Iron or Pig Iron

% of Carbon in different type of Iron

	NAME	% of C
(1)	Wrought iron	0.1 to 0.25
(2)	Steel	0.25 to 2.0
(3)	Cast Iron	2.6 to 4.3
(4)	Pig Iron	2.3 to 4.6

Process in brief :



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Manufacture of Steel:

The addition of different desired impurities into molten pure iron is known as steel making

Bessemer process :

This process involves the use of a large pear- shaped furnace (vessels) called Bessemer converter.

This is made of steel plates lined with silica (SiO_2) or magnesia (MgO) depending upon the nature of impurities. If the impurities are acidic e.g. P_4O_{10} or SiO_2, basic lining of lime (CaO) or magnesia is used (Basic process). If the impurities are basic e.g. MnO, lining of silica bricks is used (acid process). Silicon and manganese (in acidic process) or phosphorus (in basic process) are oxidised to their oxides and thus removed as slag.

Acid process : $Si + O_2 \rightarrow SiO_2$; $2Mn + O_2 \rightarrow 2MnO$; $MnO + SiO_2 \rightarrow MnSiO_3$ (Slag)

Basic process : $P_4 + 5O_2 \rightarrow P_4O_{10}$ $P_4O_{10} + 6CaO \rightarrow 2Ca_3(PO_4)_2$ (Thomas slag)

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Properties of Steel :

Steel combines the useful properties of cast iron and wrought iron. It is hard and elastic. The properties of steel depend upon its carbon content. With the increase in carbon content, the hardness of steel increases while its ductility decreases.

(i) Low carbon or soft steel contain C up to 0.25%.

(ii) Medium carbon steel or mild steel contain 0.25–0.5% C.

(iii) High carbon or hard steel contain 0.5–1.5% C.

Heat Treatment of Steel

(i) Quenching or hardening : Steel is heated to red hot temperature (700 to 800°C) and is then cooled **suddenly** by plunging into either cold water or oil. It makes steel **hard and brittle**.

(ii) Annealing : The steel is heated to red hot temperature (700 to 800°C) and then cooled slowly. It makes steel **soft**.

(iii) Tempering : If quenched steel is heated to temperature between 500 to 575 K and then cooled slowly, it becomes quite **hard but brittleness disappears**. The process is called tempering.

Surface treatment of steel

(i) Nitriding : Process of heating steel at 1000 K in an atmosphere of NH_3.
This gives hard coating of iron nitride on the surface.

(ii) Case hardening : Process of giving a thin coating of hardened steel, by heating steel in contact with charcoal followed quenching in oil.
– It is used for axles of railway wagons.

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Uses:

(i) Cast iron:

(a) It is the most important form of iron, is used for casting stoves, railway sleepers, gutter pipes, toys, etc.

(b) It is used in the manufacture of wrought iron and steel.

(ii) Wrought iron : It is used in making anchors, wires, bolts, chains and agricultural implements.

(iii) Steel :

(a) Alloy steel is obtained when other metals are added to it. Nickel steel is used for making cables, automobiles and aeroplane parts, pendulum, measuring tapes

(b) Chrome steel for cutting tools and crushing machines, and stainless steel for cycles, automobiles, utensils, pens, etc.

Extraction of Tin:

Main Ore : Cassiterite or Tinstone + (SnO_2)

Major impurities
(i) SiO_2
(ii) Sulphides of Fe & Cu
(iii) $FeWO_4 + MnWO_4$

Some following steps are involved:

(i) Crushing and concentration:

The ore is crushed and washed with a stream of running water to remove the lighter earthy and silicious impurities.

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(ii) Electromagnetic separation :

The concentrated ore is subjected to the electromagnetic separation to remove magnetic impurity of Wolframite.

(iii) Roasting :

The ore is then heated in presence of air, when volatile impurities (S as SO_2, As as As_2O_3 and Sb as Sb_2O_3) are removed. The impurities of pyrites of copper and iron are converted into their respective oxides and sulphates

$CuS + 2O_2 \rightarrow CuSO_4$; $FeS + 2O_2 \rightarrow FeSO_4$

(iv) Leaching :

Sulphates of copper and iron are dissolved in water.

(v) Washing :

The ore is washed with running water to remove the fine iron oxide produced in roasting.

Thus obtained ore contains 60 – 70% SnO_2 and is called as black tin.

(vi) Carbon reduction method : Coke & Lime stone (flux) is used.

Smelting :

The black tin is mixed with anthracite coal and heated to about 1500K in a reverberatory furnace. If SiO_2 is present as impurity then CaO is added as flux.

$SnO_2 + C \rightarrow SnO + CO$

$SnO + SiO_2 \rightarrow SnSiO_3$; $CaO + SiO_2 \rightarrow CaSiO_3$

$SnSiO_3 + CaO + C \rightarrow Sn + CaSiO_3 + CO$

or use scrap iron $SnSiO_3 + Fe \rightarrow Sn + FeSiO_3$

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(vii) Refining method :

(a) Poling

(b) Electrorefining

Anode → Impure Sn

Cathode → Pure Sn

Electrolyte → ($SnSO_4$ solution + dil. H_2SO_4)

Extraction of lead:

Main Ore : Galena (PbS) : There are mainly two types of process used in the extraction of Lead.

Ore - Galena PbS → **(a) Carbon reduction process**

Ore - Galena PbS → **(b) Self reduction process**

(a) Carbon reduction process (When impurity content is high) :

(i) Crushing & Grinding

(ii) Concentration by Forth floatation method

(iii) Roasting process with Lime stone ($CaCO_3$). Roasting in air to give PbO and SO_2 .

$2PbS(s) + 3O_2(g) \xrightarrow{\Delta} 2PbO(s) + 2SO_2$

(iv) Smelting (Carbon reduction method with coke + Fe_2O_3). Then reduction with coke or CO in a blast furnace take place.

$2PbO(s) + C \xrightarrow{\Delta} 2Pb(\ell) + CO_2(g)$

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(b) Self reduction process : (When the impurity content is less)

(i) Crushing & Grinding

(ii) Concentration by Forth floatation method

(iii) Self reduction process : PbS is partially oxidized by heating and blowing air through it. After some time the air is turned off and heating is continued. The mixture undergoes self reduction as given below.

$PbS + O_2 \longrightarrow PbO + SO_2$

$PbS + 2PbO \longrightarrow 3Pb + SO_2$

$PbS + 2O_2 \longrightarrow PbSO_4$

$PbS + PbSO_4 \longrightarrow 2Pb + 2SO_2$ } Parallel reactions

(c) Refining process :

(a) Liquation

(b) Bett's electrorefining

Anode → Impure Pb

Cathode → Pure Pb

Electrolyte → $Pb[SiF_6] + H_2SiF_6$ + Gelatine (to adjust viscosity)

on the electrolysis Pb is deposited at cathode which give 99.95% pure metal.

Extraction of Magnesium:

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(i) From Carnallite : Carnallite $KCl.MgCl_2.6H_2O$

The ore is dehydrated in a current of hydrogen chloride and the mixture of fused chlorides is electrolysed.

(ii) From Magnesite :

The concentrated ore is calcined at higher temperature

$MgCO_3 \xrightarrow[\text{Strongly}]{\text{Heated}} MgO + CO_2$

The calcined ore is heated with coke in a current of dry chlorine gas.

$MgO + C + Cl_2 \xrightarrow{\Delta} MgCl_2 + CO$

The magnesium chloride is fused and then electrolysed.

$MgO + C$ (Other reducing agents like Si, Al can be used) $\xrightarrow[\text{Vaccume}]{2000^oC} Mg + CO$

(iii) From Sea water (Dow's process) :

Sea water contains 0.13% magnesium as chloride and sulphate. It involves following steps.

(a) Precipitation of magnesium as magnesium hydroxide by slaked lime :

$MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + CaCl_2$

(b) Preparation of hexahydrated magnesium chloride

$Mg(OH)_2 + 2HCl(aq) \longrightarrow MgCl_2 + 2H_2O$

The solution on concentration and crystallisation gives the crystals of $MgCl_2.6H_2O$

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(c) Preparation of anhydrous magnesium chloride

$MgCl_2.6H_2O \xrightarrow[\text{Dry HCl(g)}]{\Delta(\text{calcination})} MgCl_2 + 6H_2O$

It is not made anhydrous by simple heating because it gets hydrolysed

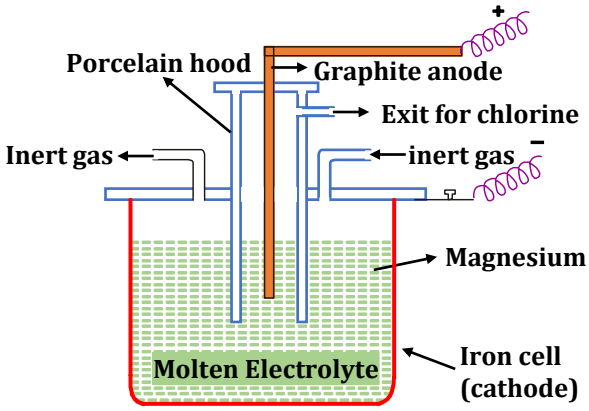
$MgCl_2.\ 6H_2O \xrightarrow{\Delta} MgO + 5H_2O + 2HCl$

(d) Electrolysis of fused anhydrous $MgCl_2$

(i) Electrolyte : Molten $MgCl_2 + NaCl + CaCl_2$

(ii) Anode : Graphite electrode

(iii) Cathode : Iron cell (steel container)



Magnesium chloride obtained by any of the above methods is fused and mixed with sodium chloride and calcium chloride in the temperature range of 973 – 1023 K. The molten mixture is electrolysed. Magnesium is liberated at the cathode (iron pot) and chlorine is evolved at graphite anode.

Reaction occurs :

The molten mixture is electrolysed. Magnesium is liberated at the cathode (iron pot) and chlorine is evolved at graphite anode.

$MgCl_2 \rightleftharpoons Mg^{2\oplus} + 2Cl^{\ominus}$

At cathode : $Mg^{2\oplus} + 2e^{\ominus} \longrightarrow Mg$(99% pure) ;

At anode : $2Cl^{\ominus} \longrightarrow Cl_2 + 2e^{\ominus}$

A stream of coal gas is passed through the pot to prevent oxidation of magnesium metal. The magnesium obtained in liquid state is purified by distillation under reduced pressure. (1 mm of Hg at 873 K).

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(iv) From Dolomite : In the Pidgeon Process Mg is Produced.

The concentrated ore is calcined at higher temperature

$CaCO_3\ .\ MgCO_3 \xrightarrow{\Delta} CaO.\ MgO + 2CO_2$

It is then reduced by **ferrosilicon alloy** at 1273 K under reduced pressure.

$2CaO.\ MgO + Fe–Si \xrightarrow{1150°C} 2Mg\uparrow + Fe + Ca_2SiO_4$

Extraction of Na:

The fused mixture of NaCl and $CaCl_2$ is taken in Down's cell which consists of circular iron cathode and carbon anode. On passing the electric current the following reactions take place :

Ionisation of NaCl	:	$NaCl \rightleftharpoons Na^{\oplus} + Cl^{\ominus}$
Collection of Na at cathode	:	$Na^{\oplus} + e^{\ominus} \rightarrow Na$(Reduction).
Collection of Cl_2 at anode	:	$Cl^{\ominus} + e^{\ominus} \rightarrow Cl$ (Oxidation), $Cl + Cl \rightarrow Cl_2$.

Extraction of silver:

Occurrence:

Ag found in free and combined state in nature.

Its main ore is Argentite Ag_2S.

Other ores are

Copper silver glance	–	Cu_2S, Ag_2S
Horn silver	–	AgCl
Argentiferrous lead	–	PbS(0.01 – 0.1% Ag)

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Steps of Metallurgy:

(1) Concentration :

(i) From Argentite (Ag_2S) As it is a sulphide ore, so froth floatation process is used.

(ii) Ag from Argentiferrous lead (PbS – Galena) : Galena has a little amount of silver. Concentration of PbS by froth floatation process

(2) Leaching and Reduction (Mac Aurther cyanide process)

(i) Formation of cyanide complex :

$$Ag_2S + 4NaCN \rightleftharpoons 2Na[Ag(CN)_2] + Na_2S \xrightarrow{+O_2+H_2O} Na_2SO_4 + 'S' + 2NaOH$$

[$\because$ O_2 is used to make reaction irreversible which remove Na_2S as $Na_2SO_4 + S$]

Na_2SO_4 does not reacts with Sodium Argento cyanide.

(ii) Displacement of Ag metal :–

$$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + \underset{(\text{black ppt})}{+ 2Ag}$$

$$Ag \xrightarrow[KNO_3]{\Delta} Ag \text{ (Solid) [Shining silver (impure)]}$$

(3) Purification of metal

(i) Parke's process : Based on distribution law.

(a) Molten Ag is more soluble in molten zinc.

Pb (molten) $\rightarrow$ Zn(molten)



(b) Melting point of Zn/Ag alloy is greater than lead, so Zn/Ag freezes first.

(c) Density of Zn/Ag is less than Pb so Zn/Ag forms the upper layer and Pb form lower layer.

(d) Upper layer taken out.

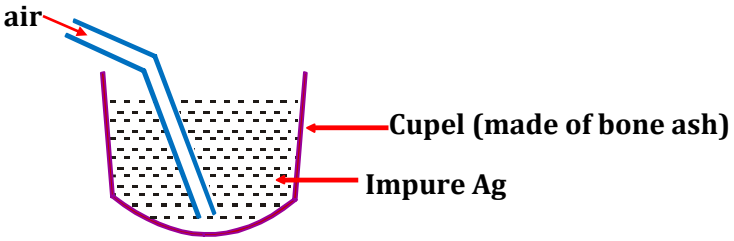
(e) Ag is separated by using distillation method. (Vapourization of Zn)

(ii) Pattinson method :–

(a) It is based upon fractional crystallisation.

(b) Ag/Pb is melted then cooled, the process is repeated again & again, concentration of Ag increases. Pb separate out in crystalline form.

(iii) Cupellation :



Pb impurities separate (oxidised) out in the form of litharge.

(iv) Electrolytic process :–

Cathode — Pure Ag

Anode — Impure Ag

Electrolyte — $AgNO_3$ + 1% HNO_3 (To increase ionisation and to avoid hydrolysis of $AgNO_3$ by common ion effect)

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Uses:

(i) It is used in silver plating.

(ii) Silver foils are used in medicine.

(iii) Silver amalgam is used for dental filling.

(iv) Compounds of silver are used in silvering of mirrors ($AgNO_3$ + HCHO + Red Pb), in photography, as laboratory reagents etc.

(v) Silver is easily alloyed with copper, so it is used in making coins, ornaments, silver ware etc.

(vi) It gives black spot on skin due to decomposition so it is also used as hair dye and ink.

Extraction of Gold :

Occurrence : Au found in free (native) state in nature.

Same steps as are involved in the extraction of Silver metal.

(i) Crushing & Grinding :

(ii) Leaching process : Gold are extracted by the cyanide process (Mc Arthur - Forest process).

Reaction involved :

$$4Au + 8NaCN + O_2 + 2H_2O \rightleftharpoons 4Na[Au(CN)_2] + 4NaOH$$

$$2Na[Au(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Au$$

(iii) Refining Process:

Anode → Impure Au

Cathode → Pure Au

Electrolyte → ($AuCl_3$ solution + dil. HCl)

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Aluminium	1. Bauxite, Al_2O_3. x H_2O 2. Cryolite, Na_3AlF_6	Electrolysis of Al_2O_3 dissolved in molten Na_3AlF_6	For the extraction, a good source of electricity is required.
Iron	1. Haematite, Fe_2O_3 2. Magnetite, Fe_3O_4	Reduction of the oxide with CO and coke in Blast furnace	Temperature approaching 2170 K is required.
Copper	1. Copper pyrites, $CuFeS_2$ 2. Copper glance, Cu_2S 3. Malachite, $CuCO_3.Cu(OH)_2$ 4. Cuprite, Cu_2O	Roasting of sulphide partially and reduction	It is self reduction in a specially designed converter. The reduction takes place easily. Sulphuric acid leaching is also used in hydrometallurgy from low grade ores.
Zinc	1. Zinc blende or Sphalerite, ZnS 2. Calamine, $ZnCO_3$ 3. Zincite, ZnO	Roasting followed by reduction with coke	The metal may be purified by fractional distillation.

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